



- Drove data-driven project management by deploying analytics tooling to surface velocity, defect density, and milestone risk metrics to stakeholders.
- Managed resource allocation and risk mitigation across competing program milestones, acting as the primary interface between software, hardware, and systems engineering teams.

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### **Team Leader & ECEMTAG**

November 2017 - May 2021 & Istanbul, TR

Led design and validation of model-based control algorithms for ECUs and TCUs across automotive and defense programs.

- Introduced automated testing frameworks and HIL simulation environments that raised software verification coverage and reduced physical prototype dependency.
- Developed safety and redundancy functions in compliance with military standards, coordinating requirements directly with defense customers.
- Acted as cross-functional integration lead, aligning software and hardware milestones across embedded, systems, and validation teams using JIRA-based program tracking.
- Contributed to E/E architecture process development and rollout, including tool qualification and development workflow standardization.

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### **Lead Researcher & TUBITAK (Scientific and Technological Research Council of Turkey)**

December 2016 - November 2017 & Kocaeli, TR

- Directed automotive and defense research programs focused on advanced control systems and diagnostics.
- Developed a product-level tool for data dictionary generation, calibration, and MiL (Model-in-Loop) testing, accelerating algorithm validation workflows.
- Characterized IPMSM performance in motoring and regenerative modes on a dynamometer test bench.

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### **Senior Systems Engineer & TUMOSAN**

January 2016 - December 2016 & Istanbul, TR

- Developed automatic shift-selection logic using Simulink Stateflow for a hybrid drivetrain platform.
- Designed traction control algorithms for slip-slide protection under variable surface conditions.
- Optimized fuel consumption and battery usage profiles using multi-objective cost functions.

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### **Systems Engineer & METRO CO**

February 2013 - January 2016 & Kocaeli, TR

- Built a longitudinal vehicle model and derived drivetrain power and torque requirements from first principles.
- Developed a full PMSM driver including Maximum Torque Per Ampere (MTPA) and flux-weakening control algorithms.

- Automated PID tuning via phase and gain margin calculation, and conducted HIL validation using dSpace MABX-I.

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Researcher && Iowa State University - Bio-Electric Laboratory

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- Supervised advanced research on transcranial magnetic stimulation (TMS) and electric field modeling.
- Instructed electric machines coursework to sophomore engineering students.

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